A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Circular Motion

Anything moving in a circle is accelerating, even if its speed never changes, because its direction is changing all the time. That acceleration points at the centre of the circle, and it needs a resultant force pointing at the centre too. In this topic we measure angles in radians, describe spinning with angular velocity, derive the centripetal acceleration from a vector diagram, and then use $F = mv^2/r$ for cars on bends, balls on strings and roller coasters.

Measuring angles in radians

Degrees are a human choice: someone decided a full turn should be 360. Physicists prefer an angle unit that comes from the circle itself. Take a circle of radius r, walk a distance s round the edge, and the angle you have turned through at the centre is

$$\theta = \frac{s}{r} \qquad (\text{in radians})$$

DEFINITION

The radian

One **radian** is the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle.

So an angle of 1 rad means "the arc is one radius long". A full turn has an arc of one circumference, $2\pi r$, so a full turn is $2\pi r/r = 2\pi$ radians. That gives the conversion:

$$2\pi \text{ rad} = 360^\circ \qquad \Rightarrow \qquad 1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57.3^\circ$$

- **Degrees to radians:** multiply by $\dfrac{\pi}{180}$.
- **Radians to degrees:** multiply by $\dfrac{180}{\pi}$.

Degrees	30°	45°	60°	90°	180°	270°	360°
Radians (exact)	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$	2π
Radians (decimal)	0.524	0.785	1.05	1.57	3.14	4.71	6.28

EXAM TIP

Check your calculator mode. In circular motion questions, use **radians** for anything involving ω and $\theta = s/r$.

Because the radian is a ratio of two lengths it has no real dimensions, which is why ω in rad s^{-1} can be written as just s^{-1} in a units check.

Angular velocity

For something going round, "how fast" can mean two things. A horse near the edge of a carousel covers more metres per second than one near the middle, but both go round once in the same time. **Angular velocity** captures the second idea: how quickly the angle changes.

DEFINITION

Angular velocity

The **angular velocity** ω of an object moving in a circle is the angle swept out by the radius per unit time, $\omega = \dfrac{\Delta\theta}{\Delta t}$. It is measured in rad s^{-1}.

In one complete revolution the angle is 2π and the time taken is the period T, so $\omega = 2\pi/T$. Rearranged, this is the formula-sheet version. Since frequency $f = 1/T$, you can also write $\omega = 2\pi f$.

Period

FORMULA SHEET

$$T = \frac{2\pi}{\omega}$$

T: time for one revolution / s

ω: angular velocity / rad s^{-1}

Linear speed

FORMULA SHEET

$$v = \omega r$$

v: speed along the circle / m s^{-1} r: radius / m

Where v = ωr comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. In a short time Δt the object turns through an angle $\Delta\theta$. By the definition of angular velocity:

$$\Delta\theta = \omega\,\Delta t$$

2. The distance travelled along the circle is the arc length. From the definition of the radian, $s = r\theta$:

$$\Delta s = r\,\Delta\theta = r\,\omega\,\Delta t$$

3 Speed is distance divided by time:

$$v = \frac{\Delta s}{\Delta t} = \frac{r\,\omega\,\Delta t}{\Delta t} \quad \Rightarrow \quad \boxed{v = \omega r}$$

This only works with ω in radians per second, because $s = r\theta$ is only true in radians.

KEY IDEA

Same ω, different v

Every point on a rigid spinning object (a wheel, a disc, the Earth) has the **same angular velocity**. Its speed grows in proportion to its distance from the axis, $v = \omega r$. Twice as far out means twice as fast.

WORKED EXAMPLE

The Earth turns once every 24 hours. Its radius is 6.37×10^6 m. Calculate (a) the Earth's angular velocity, (b) the speed of a person on the equator, and (c) the speed of a person in London, at latitude 51.5°.

Show solution

(a) Convert the period to seconds: $T = 24 \times 3600 = 86\,400$ s.

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{86\,400} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$$

(b) On the equator the radius of the circle is the Earth's radius:

$$v = \omega r = 7.27 \times 10^{-5} \times 6.37 \times 10^6 = 463 \text{ m s}^{-1}$$

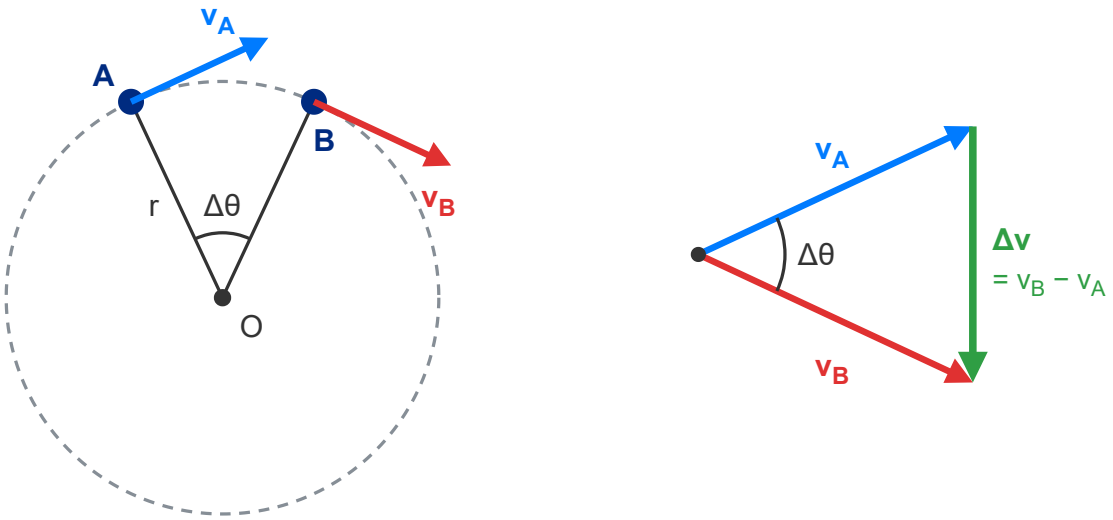
(c) In London you travel round a smaller circle, whose radius is the distance from the Earth's axis: $r = R\cos 51.5^\circ = 6.37 \times 10^6 \times 0.623 = 3.97 \times 10^6$ m. The angular velocity is the same:

$$v = 7.27 \times 10^{-5} \times 3.97 \times 10^6 = 288 \text{ m s}^{-1}$$

Both of you go round once a day (same ω), but the person on the equator has further to go.

Centripetal acceleration

Velocity is a vector. Even when the speed is constant, the **direction** of the velocity changes as an object goes round a circle, so the velocity changes, so the object is accelerating. Which way? The best way to find out is to draw the velocity at two nearby moments and look at the difference.



Left: the object moves from A to B at constant speed. Right: the same two velocity vectors drawn from one point. Δv joins the tip of v_A to the tip of v_B, and it points straight down the page, towards the centre of the circle.

Deriving a = v²/r with a vector diagram

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 An object moves round a circle of radius r at constant speed v. In a short time Δt it moves from A to B, and the radius turns through a small angle $\Delta\theta$. The velocity is always at 90° to the radius, so the velocity also turns through $\Delta\theta$:

$$|v_A| = |v_B| = v, \qquad \text{angle between } v_A \text{ and } v_B = \Delta\theta$$

2 The change in velocity is $\Delta v = v_B - v_A$. Draw v_A and v_B tail to tail: Δv is the arrow from the tip of v_A to the tip of v_B. This is an isosceles triangle with two sides of length v and the angle $\Delta\theta$ between them.

By symmetry, Δv is perpendicular to the line halfway between v_A and v_B, so it points along the radius at the midpoint of the arc, towards O.

3 When $\Delta\theta$ is small, the short side of the triangle is almost the same as an arc of a circle of radius v. Arc length = radius × angle, so

$$|\Delta v| \approx v\,\Delta\theta$$

4 Acceleration is change in velocity over time:

$$a = \frac{|\Delta v|}{\Delta t} = v\,\frac{\Delta\theta}{\Delta t} = v\,\omega$$

$\Delta\theta/\Delta t$ is the angular velocity ω.

5 From $v = \omega r$, $\omega = v/r$. Substitute:

$$a = v \times \frac{v}{r} \quad \Rightarrow \quad \boxed{a = \frac{v^2}{r}}$$

As $\Delta t \to 0$, A and B merge, the approximation in step 3 becomes exact, and Δv points exactly at the centre. The acceleration is directed **towards the centre** of the circle: it is **centripetal** ("centre-seeking").

Deriving a = rω² SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Start from the result we just derived:

$$a = \frac{v^2}{r}$$

2 Substitute $v = \omega r$:

$$a = \frac{(\omega r)^2}{r} = \frac{\omega^2 r^2}{r} \quad \Rightarrow \quad \boxed{a = r\omega^2}$$

Step 4 of the previous derivation also gave the handy middle form $a = v\omega$.

Centripetal acceleration FORMULA SHEET

$$a = \frac{v^2}{r}$$

a: acceleration towards centre / m s^{-2}

Centripetal acceleration FORMULA SHEET

$$a = r\omega^2$$

ω: angular velocity / rad s^{-1}

COMMON MISTAKE

"Constant speed, so no acceleration" is the classic wrong answer. In uniform circular motion the **speed** is constant but the **velocity** is not, so there is an acceleration, of size v^2/r, always pointing at the centre.

WORKED EXAMPLE

A washing machine drum of radius 0.25 m spins at 1200 revolutions per minute. Calculate the speed of a sock pressed against the drum wall and its centripetal acceleration. How many times larger than g is this?

Show solution

Convert rpm to rad s⁻¹: each revolution is 2π rad, and there are 60 s in a minute.

$$\omega = \frac{1200 \times 2\pi}{60} = 126 \text{ rad s}^{-1}$$

$$v = \omega r = 126 \times 0.25 = 31.4 \text{ m s}^{-1}$$

$$a = r\omega^2 = 0.25 \times 126^2 = 3.9 \times 10^3 \text{ m s}^{-2} \approx 400\,g$$

The drum wall pushes the sock towards the centre with a huge force. Water in the sock is not held by the wall at the holes, so it carries on in a straight line, out through the holes.

Centripetal force

Newton's first law says an object keeps moving in a straight line at constant speed unless a resultant force acts on it. So to keep something going round a circle, there must be a **resultant force** on it, and by Newton's second law it points the same way as the acceleration: **towards the centre**.

Centripetal force

FORMULA SHEET

$$F = ma = \frac{mv^2}{r}$$

F: resultant force towards centre / N

m: mass / kg

Centripetal force

FORMULA SHEET

$$F = mr\omega^2$$

r: radius / m ω: angular velocity / rad s⁻¹

DEFINITION

Centripetal force

The **centripetal force** is the **resultant force** on an object moving in a circle. It is directed towards the centre of the circle, and it is needed both to **produce** circular motion and to **maintain** it.

KEY IDEA

Centripetal force is a job, not a new force

"Centripetal" describes the direction of a resultant force. Something real has to supply it, and you should never draw an extra arrow labelled "centripetal force" on a free-body diagram.

Situation	What provides the centripetal force
Puck on a string on an air table	Tension in the string
Car going round a flat bend	Friction from the road on the tyres
The Moon orbiting the Earth	Gravitational force from the Earth
Clothes in a spin dryer	Normal contact force from the drum wall
Electron in a uniform magnetic field	Magnetic force Bqv
Electron round a nucleus (simple model)	Electric (Coulomb) attraction

Because the centripetal force is always at 90° to the velocity, it never pushes the object forwards or backwards along its path. It does **no work**, so it changes the direction of motion but not the speed or the kinetic energy.

COMMON MISTAKE

There is no outward force

When a car turns left you feel "thrown" to the right. Nothing is pushing you right. Your body is trying to carry on in a straight line (Newton's first law) while the car turns underneath you, until the door pushes you **inwards**, towards the centre of the turn. If the string breaks, a ball flies off along the tangent, not radially outwards.

WORKED EXAMPLE

A 0.20 kg puck on a frictionless table is whirled in a horizontal circle of radius 0.50 m on a string, making 2.0 revolutions per second. (a) Calculate the tension in the string. (b) The string snaps at 25 N. What is the greatest number of revolutions per second it can make?

Show solution

(a) $\omega = 2\pi f = 2\pi \times 2.0 = 12.6 \text{ rad s}^{-1}$. The tension is the only horizontal force, so it is the centripetal force:

$$T = mr\omega^2 = 0.20 \times 0.50 \times 12.6^2 = 15.8 \text{ N}$$

(b) Rearrange for ω with $T = 25$ N:

$$\omega = \sqrt{\frac{T}{mr}} = \sqrt{\frac{25}{0.20 \times 0.50}} = 15.8 \text{ rad s}^{-1}, \qquad f = \frac{\omega}{2\pi} = 2.5 \text{ revolutions per second}$$

Cars on bends and banked tracks

On a flat road, the only horizontal force on a car is **friction** from the road on the tyres, pointing towards the centre of the bend. The friction needed is

$$F = \frac{mv^2}{r}$$

The v^2 is the important part. Double your speed round a bend and you need **four times** as much friction. A tighter bend (smaller r) also needs more. The road can only supply so much grip, and it supplies much less when it's wet or icy. If the friction needed is more than the road can give, the car cannot follow the bend: it moves on a wider curve, closer to a straight line, and slides towards the outside of the bend.

WORKED EXAMPLE

A car of mass 1200 kg goes round a flat bend of radius 50 m at 15 m s^{-1}. (a) What friction force is needed? (b) The greatest friction the road can provide is 7200 N. What is the maximum safe speed?

Show solution

$$F = \frac{mv^2}{r} = \frac{1200 \times 15^2}{50} = 5400 \text{ N}$$

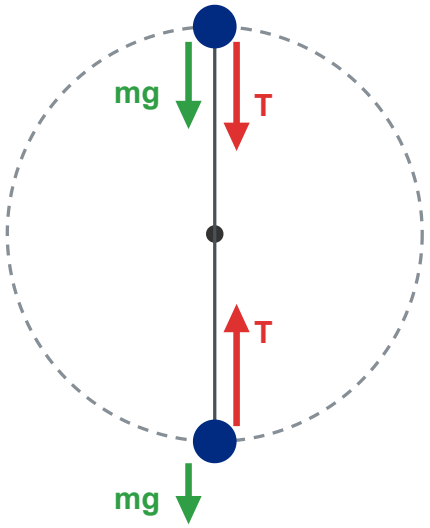
(b) Rearrange with $F = 7200$ N:

$$v = \sqrt{\frac{Fr}{m}} = \sqrt{\frac{7200 \times 50}{1200}} = 17.3 \text{ m s}^{-1}$$

Only 15% faster than before uses up all the grip, because $F \propto v^2$: $1.15^2 \approx 1.33$, and $5400 \times 1.33 = 7200$.

Vertical circles

Now swing a ball on a string in a **vertical** circle, or ride a roller coaster round a loop. Nothing new is needed: at every point, the resultant force towards the centre must equal mv^2/r. What changes is that the weight now sometimes points towards the centre and sometimes away from it, and the speed changes as the ball rises and falls (kinetic energy is traded for gravitational potential energy).



At the top

both forces point to the centre:

T + mg = mv²/r

At the bottom

weight points away from the centre:

T − mg = mv²/r

Free-body diagrams at the top and bottom of a vertical circle. Take "towards the centre" as positive.

- **At the bottom:** $T = mg + \dfrac{mv^2}{r}$. The string has to hold the ball up *and* provide the centripetal force, so the tension is greatest here. This is where a string is most likely to snap, and where you feel heaviest on a ride.
- **At the top:** $T = \dfrac{mv^2}{r} - mg$. Gravity is already helping, so less tension is needed. The ball is also moving slowest here.
- **Minimum speed at the top:** the tension can't be negative (a string can pull but not push). The smallest possible speed is when $T = 0$ and gravity alone provides the centripetal force: $mg = mv^2/r$, so $v_{\min} = \sqrt{gr}$. Any slower, and the string goes slack and the ball falls inside the circle as a projectile.

The same reasoning explains why water stays in a bucket swung over your head: if $v \geq \sqrt{gr}$ at the top, the bucket has to push the water down to keep it on the circle, so the water stays pressed into the bucket.

WORKED EXAMPLE

A roller coaster car (total mass 500 kg) enters a vertical loop of radius 8.0 m at 22 m s⁻¹ at the bottom. Ignore friction. (a) What is the minimum speed needed at the top? (b) Find the speed at the top. (c) Find the normal contact force from the track on the car at the top. (d) What force does the seat exert on a 60 kg rider at the bottom?

Show solution

(a) At the minimum speed the track exerts no force and weight alone provides the centripetal force:

$$v_{\min} = \sqrt{gr} = \sqrt{9.81 \times 8.0} = 8.86 \text{ m s}^{-1}$$

(b) The top is $2r = 16$ m above the bottom. Energy is conserved, so $\tfrac{1}{2}mv_{\text{top}}^2 = \tfrac{1}{2}mv_{\text{bottom}}^2 - mg(2r)$. Cancel m:

$$v_{\text{top}}^2 = 22^2 - 2 \times 9.81 \times 16 = 484 - 314 = 170, \qquad v_{\text{top}} = 13.0 \text{ m s}^{-1}$$

(c) At the top, the normal force N and the weight both point down, towards the centre:

$$N + mg = \frac{mv^2}{r} \quad \Rightarrow \quad N = \frac{500 \times 170}{8.0} - 500 \times 9.81 = 10\,600 - 4900 = 5.7 \times 10^3 \text{ N}$$

(d) At the bottom the seat pushes up and the weight acts down:

$$N = mg + \frac{mv^2}{r} = 60 \times 9.81 + \frac{60 \times 22^2}{8.0} = 589 + 3630 = 4.2 \times 10^3 \text{ N}$$

EXAM TIP

For every circular motion question, use the same three steps: (1) draw the real forces, (2) find the **resultant towards the centre**, (3) set it equal to mv^2/r or $mr\omega^2$. If the speed changes (vertical circles), use conservation of energy to find v at the point you need.

Quick checks

QUICK CHECK

What is 1.5 rad in degrees?

- 27°
- 86°
- 270°
- 0.026°

$1.5 \times \frac{180}{\pi} = 85.9°$. Quick check: 1 rad is about 57°, so 1.5 rad is a bit less than 90°. 0.026 is what you get converting the wrong way.

QUICK CHECK

A car drives round a roundabout at a constant speed. Which statement is correct?

- Its velocity is constant.
- Its acceleration is zero.
- Its acceleration is directed towards the centre of the roundabout.
- The resultant force on it is directed away from the centre.

The direction of the velocity keeps changing, so the car accelerates, towards the centre, with $a = v^2/r$. The resultant force points the same way as the acceleration: inwards.

QUICK CHECK

Two children sit on a spinning roundabout, one 1.0 m from the centre and one 2.0 m from the centre. Compared with the inner child, the outer child has

- the same angular velocity and twice the speed.
- twice the angular velocity and the same speed.
- the same speed and the same acceleration.
- half the centripetal acceleration.

They go round together, so ω is the same. $v = \omega r$ doubles, and $a = r\omega^2$ also doubles (not halves).

QUICK CHECK

A car goes round a bend of fixed radius. If its speed doubles, the friction force needed

- doubles.
- becomes four times bigger.
- halves.
- stays the same.

$F = mv^2/r$, so $F \propto v^2$: $2^2 = 4$.

QUICK CHECK

A ball on a string is swung in a vertical circle. Where is the tension in the string greatest?

- At the top
- At the bottom
- Halfway up the sides
- It is the same everywhere

At the bottom the ball is fastest, so it needs the most centripetal force, and the string must also cancel the weight: $T = mg + mv^2/r$.

QUICK CHECK

A puck whirled in a horizontal circle on a frictionless table is moving due north when the string breaks. Which way does it move next?

- Radially outwards, away from the centre
- Due north, in a straight line
- In a curve that gradually straightens
- Towards the centre

Once the string breaks there is no horizontal force, so by Newton's first law it carries on in a straight line with the velocity it had at that instant, which was along the tangent.